

AI3002 – Search Methods in AI

Travelling Salesman Problem

2025-09-26

Given a set of N cities and the distance between every pair of cities, find the shortest possible route that starts from a city and visits every city exactly once and returns to the starting city. Write a program to find as short a tour as possible of the N cities.

This assignment carries 7% of your course grade. The assignment has to be done in teams of 3. Only one member from each team needs to submit the assignment.

1. Input Description

Your program must read from a file with the following structure:

- **First line:** either EUCLIDEAN or NON-EUCLIDEAN.
- **Second line:** an integer N , number of cities.
- **Next N lines:** each line contains N distance values, i.e., a_{ij} is distance between i^{th} and j^{th} node.

The distance matrix contains floating point numbers. Use the distance matrix for computing tour cost.

For example, a Euclidean TSP for $N = 5$ might have the following input:

```
1 EUCLIDEAN
2 5
3 0.0 4.597411 9.110738 3.187861 4.302992
4 4.597411 0.0 13.285327 5.736168 4.420019
5 9.110738 13.285327 0.0 7.822640 10.096052
6 3.187861 5.736168 7.822640 0.0 2.419287
7 4.302992 4.420019 10.096052 2.419287 0.0
```

Similarly, a Non-Euclidean TSP for $N = 3$ might have the following input:

```
1 NON-EUCLIDEAN
2 3
3 0.0 100.0 2.0
4 100.0 0.0 5.0
5 2.0 5.0 0.0
```

Note:

1. For N nodes, the nodes are labelled from 0 to $N - 1$.
2. Distance matrix represents an undirected fully connected graph. The diagonal elements are 0 and every non-diagonal element has a finite positive value.

Sample input files are uploaded for your reference.

2. Output Description

Each line of your output should contain a **valid** tour in **path representation**. You should keep writing to the output file as you discover better solutions (with lower costs). The **last line** of the output will be used for evaluation. If you do not output any tours, it will lead to a score of 0 in that test case.

For $N = 5$, following is a valid output file -

```
1 0 2 1 3 4
2 1 2 4 3 0
3 4 3 1 2 0
```

In this case, the tour 4 3 1 2 0 will be used for evaluation.

3. Evaluation

There will be 6 test cases -

1. EUCLEDIAN ($N = 50$)
2. EUCLEDIAN ($N = 100$)
3. EUCLEDIAN ($N = 200$)
4. NON-EUCLEDIAN ($N = 50$)
5. NON-EUCLEDIAN ($N = 100$)
6. NON-EUCLEDIAN ($N = 200$)

Your program will be run for each test case with a timeout of 300s. We will terminate your program after this limit (if you do not exit early). The last line of your output file will be used for evaluation.

Grading is relative, the best tour will get the highest points. . The team with the highest cost for a test case will get the lowest score.

4. Final Submission Details (6%)

4.1. Code Submission

You are expected to submit a .zip file with the main.py at the root. Auxiliary/helper files are allowed and you should import them in your main.py. You are only allowed to use Python3.

The auto-grader will run your code with the following format -

```
1 python3 main.py /path/to/input.txt /path/to/output.txt
```

1. Make sure your code runs on a Linux environment.
2. You will be allowed to use **NumPy** for any matrix operations. You are not allowed to use any other **external libraries**.
3. If your program has any execution or runtime errors for the aforementioned input/output formats, zero marks will be awarded.

Your .zip file should be named `id1_id2_id3.zip`. Example - `U20230001_U20230013_U20230134.zip`

4.2. Report Submission

Your code submission should be accompanied with a handwritten report (upto 2 A4 pages) detailing your methodology, experiments, and approach. This report is not graded but is **mandatory** for the code submission. Failing to submit the report or writing the report in an incomprehensible manner will lead to a 0 grade in the assignment.

5. Trial Submission Details (1%)

You can choose to make a trial code submission on or before **October 6, 2025**. Your code submission should be in the same format as above, and should execute successfully (with at least 1 valid tour within the time limit). This will allow you to check if your code runs successfully on our Linux environment. A successful run for our test cases will give you the entire 1%.

Note - Your tours will not be ranked and do not carry any grade. Only successful execution does.